Roken

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Abstract
Roken - recurrent Noken, a dynamic implicit deep learning model of attention, designed to integrate state tracking into large language model next-token and sequence-to-sequence learning.

Introduction

The implicitRNN model [1] functions similarly to a vanilla RNN, processing sequence inputs one by one. For each time step i, where $t_i \in R^p$ represents the i-th element in a sequence $(t_1, t_2, \cdots, t_m)$. The model input u_i for implicitRNN is a concatenation of t_i and the previous hidden state h_{i-1}, akin to a vanilla RNN. The implicit prediction rule for implicitRNN is as follows:

$h_0 = 0$, $s_0 = 0$

$u_i = [t_i \; h_{i-1}]'$

$s_i = \phi\,(A s_i + B u_i)$ (equilibrium equation)

$y_i(u_i) = C s_i + D u_i$ (prediction equation)

$h_i = y_i(u_i)$

In this setup, the recurrent layer is replaced with an implicit structure comprising the equilibrium and prediction equations, where we learn [2] matrices A, B, C, D and fixed points s_i.

Let score be a desired similarity function, like a reproducing kernel for example. Input u_i corresponding to token t_i, generates a unique fixed point s_i via the equilibrium equation. The hidden states h, get updated via the prediction equation. Akin to transformer attention looking at query-key similarity, fix $i \in [m]$, normalize score(h_i, s_j) into probabilities, for hidden state h_i and fixed points s_j, $j \in [i]$, to create a probability distribution over (latent) tokens, and output the most likely token. LLMs typically minimize perplexity, after applying layers of feedforward networks to the h_i. The wrinkle here is we can also regularize with the average Renyi α-entropy of these distributions. This enables robust next-token learning. Moreover, if we think of the latent tokens s_j as possible meanings of the hidden state h_i, then lowering average Renyi α-entropy sharpens and aligns semantics. Similar considerations (for score(h_i, h'_j), state h_i corresponding to target token t_i and state h'_j corresponding to source token t'_j) apply to sequence-to-sequence translation.

Note that perplexity attends to syntactic co-occurrence, while average Renyi α-entropy attends to semantic alignment. The objective of a token learning problem can be tailored to any desired emphasis between the two by regularization.

Inputs:
Target tokens $t1, t2, \dots, tm$
Source tokens $t'1, t'2, \dots, t'm'$

State Machine
Initial state $h0 = 0$
Next state $hi = S(hi\text{-}1, ti)$, state function S
Output $oi{+}1 = O(hi)$, output function O

ImplicitRNN
$h0 = 0$, $s0 = 0$
$ui = [ti\ hi\text{-}1]$
$si = \phi(Asi + Bui)$
$yi(ui) = Csi + Dui$
$hi = yi(ui) =: S(hi\text{-}1, ti)$

Update each hi by applying an L-layer feedforward network with parameters θ, and in the following assume that this has been done.

Given input token ti, define output
$oi{+}1 = tj^*(i) =: O(hi)$
where βij = soft max over $j \varepsilon [i]$ score(hi, sj)
and j^*i = arg max over $j \varepsilon [i]$ βij
and score is a desired similarity function, like a reproducing kernel.

Define the average Renyi α-entropy [3]
$E(\{ti\}) = (1/m)$ Sum over i, $H(i,\alpha)$,
where $H(i,\alpha) = H\alpha(\beta ij)$, the Renyi α-entropy of $\{\beta ij\}$.

The next-token learning problem:
Let $\lambda > 0$, learn matrices A, B, C, D and fixed points si, and FFN parameters θ, to minimize
$\{$ Perplexity$(o2, \dots, om{+}1) + \lambda E(\{ti\})$ $\}$

That is minimize perplexity regularized by average Renyi α-entropy, to ensure syntactic and semantic fidelity.

Next we consider sequence to sequence learning:
Solve the next-token problems for {ti} and {t'j}, to learn FFN updated {hi} and {h'j}.
For i in [m], define δij = soft max over j in [m'], score (hi, h'j)
and define j*i = arg max over j, δij.
Given target input ti, define source output oi = t'j*(i),
so t1, t2, ... tm is translated into o1, o2, ... , om, oi ε {t'j}.

That is learn (minimize perplexity regularized by average Renyi α-entropy) hidden states of implicit models for target and source tokens, then translate the target token into the source token corresponding to the most similar source state to the target state.

Discussion

Dynamic state tracking [4], the iterative updating of latent variables reflecting evolving context, involves sequential dependencies that transformer feedforward network layer topologies struggle to maintain. To address this limitation we propose an update to recurrent implicit dynamic LLM [3]. Moreover, low displacement rank operators and robust implicit via non-Euclidean contractions [2] ensure fast tractable chunk scaling via mini-batch SGD.

Roken - a recurrent implicit dynamic LLM

State Machine

Input: target tokens $t_1, \ldots, t_m \in \mathbb{R}^p$

Initial state: $h_0 = 0$

Next state: $h_i = \mathcal{S}(h_{i-1}, t_i)$

Output: $o_{i+1} = \mathcal{O}(h_i)$, $i \in [m]$

Implicit RNN [1]

$$h_0 = 0 \in \mathbb{R}^q, \quad s_0 = 0 \in \mathbb{R}^n$$

$$u_i = \begin{pmatrix} t_i \\ h_{i-1} \end{pmatrix} \in \mathbb{R}^r \quad (p+q=r)$$

$$s_i = \varphi(\underset{n\times n}{A} s_i + \underset{n\times r}{B} u_i) \in \mathbb{R}^n$$

$$\hat{y}_i(u_i) = \underset{q\times n}{C} s_i + \underset{q\times r}{D} u_i \in \mathbb{R}^q$$

$$h_i = \hat{y}_i(u_i) =: \mathcal{S}(h_{i-1}, t_i) \in \mathbb{R}^q$$

where $A = J - \mathrm{diag}(|J| \mathbb{1}_n) + \gamma I_n$, $0 < \gamma < 1$,

makes A well-posed for φ. [2]

For L feedforward network layers

let $\Theta = \{ (\underset{q \times q}{W_l}, \underset{q \times 1}{b_l}) \}_{l=1}^{L}$

Define $h_{i,0} = h_i$,

and for $l \in [L]$, $h_{i,l} = \mathrm{ReLu}(W_l h_{i,l-1} + b_l)$

Define $\beta_{ij} = \underset{j \in [i]}{\mathrm{softmax}}\ \mathrm{score}(h_{i,L}, s_j)$

and $j_i^{\beta} = \underset{j \in [i]}{\arg\max}\ \beta_{ij}$

where score is a similarity function, like a kernel.

Given input token t_i, update h_i to $h_{i,L}$

and define output $o_{i+1} = \Phi(h_i) \triangleq t_{j_i^{\beta}}$

Define the average Renyi α-entropy of $\{t_i\}_{i=1}^m$

by: $E(\{t_i\}_{i=1}^m) = \frac{1}{m}\sum_{i=1}^m H_\alpha^i$,

where $H_\alpha^i \triangleq H_\alpha(\{\beta_{ij}\}_{j=1}^i)$

and H_α is Renyi α-entropy

Next-token learning problem*

Let $\lambda > 0$, learn matrices A, B, C, D and

fixed points s_i and FFN parameters Θ to minimize:

$$\{ \text{Perplexity}(o_2, \ldots, o_{m+1}) + \lambda E(\{t_i\}_{i=1}^m) \}$$

Next-token prediction

Initialization; let t_1 = the empty token or a seed token

Given $[t_1]$, define $t_2 = o_2$

Given $[t_1, t_2]$, define $t_3 = o_3$, and so on,

autoregressively generating the next token continuation.

Sequence to sequence learning*

Input: target tokens $t_1, \ldots, t_m$

source tokens $t'_1, \ldots, t'_{m'}$

For $\{t_i\}$ learn A, B, C, D and s_i and θ to minimize

$$\{ \text{Perplexity}(o_2, \ldots, o_{m+1}) + \lambda E(\{t_i\}_{i=1}^{m}) \}$$

Similarly, For $\{t'_j\}$ learn A', B', C', D' and s'_j

and θ' to minimize $\{ \text{Perplexity}(o'_2, \ldots, o'_{m'+1}) + \lambda' E(\{t'_j\}_{j=1}^{m'}) \}$

Consequently we know $\{h_{i,L}\}_{i=1}^{m}$ and $\{h'_{j,L'}\}_{j=1}^{m'}$

Define $\delta_{ij} = \underset{j \in [m']}{\text{softmax}} \; \text{score}(h_{i,L}, h'_{j,L'})$, $i \in [m]$

and $j_i^{\delta} = \underset{j \in [m']}{\arg\max} \; \delta_{ij}$

Given input t_i, $i \in [m]$, define output $o_i = t'_{j_i^{\delta}}$

So $t_1, \ldots, t_m$ is translated to $o_1, \ldots, o_m$, $o_i \in \{t'_j\}_{j=1}^{m'}$

* In the learning problems, the number of input tokens is the mini-batch size for stochastic gradient descent. On each corpus chunk, we use mini-batch SGD to jointly learn embeddings A, B, C and D, fixed points s_i, hidden states h_i, and feedforward network parameters θ, so as to minimize perplexity regularized by average Renyi α-entropy.

References

[1] 'The Extrapolation Power of Implicit Models'
Juliette Decugis, Alicia Y. Tsai,
Max Emerling, Ashwin Ganesh, Laurent El Ghaoui
arXiv:2407.14430v1 [cs.LG] 19 Jul 2024

[2] 'Robust Implicit Networks via Non-Euclidean Contractions'
Saber Jafarpour, Alexander Davydov,
Anton V. Proskurnikov, Francesco Bullo
arXiv:2106.03194v6 [cs.LG] 26 Jan 2022

[3] 'Implicitly learning token semantics'
Gary Nan Tie, 3 Mar 2026
DOI: 10.13140/RG.2.2.30432.34566/1

[4] 'The Topological Trouble with Transformers'
Michael C. Mozer, Shaib Ahmed Siddiqui, Rosanne Liu
arXiv:2604.17121v3 [cs.LG] 3 Jun 2026